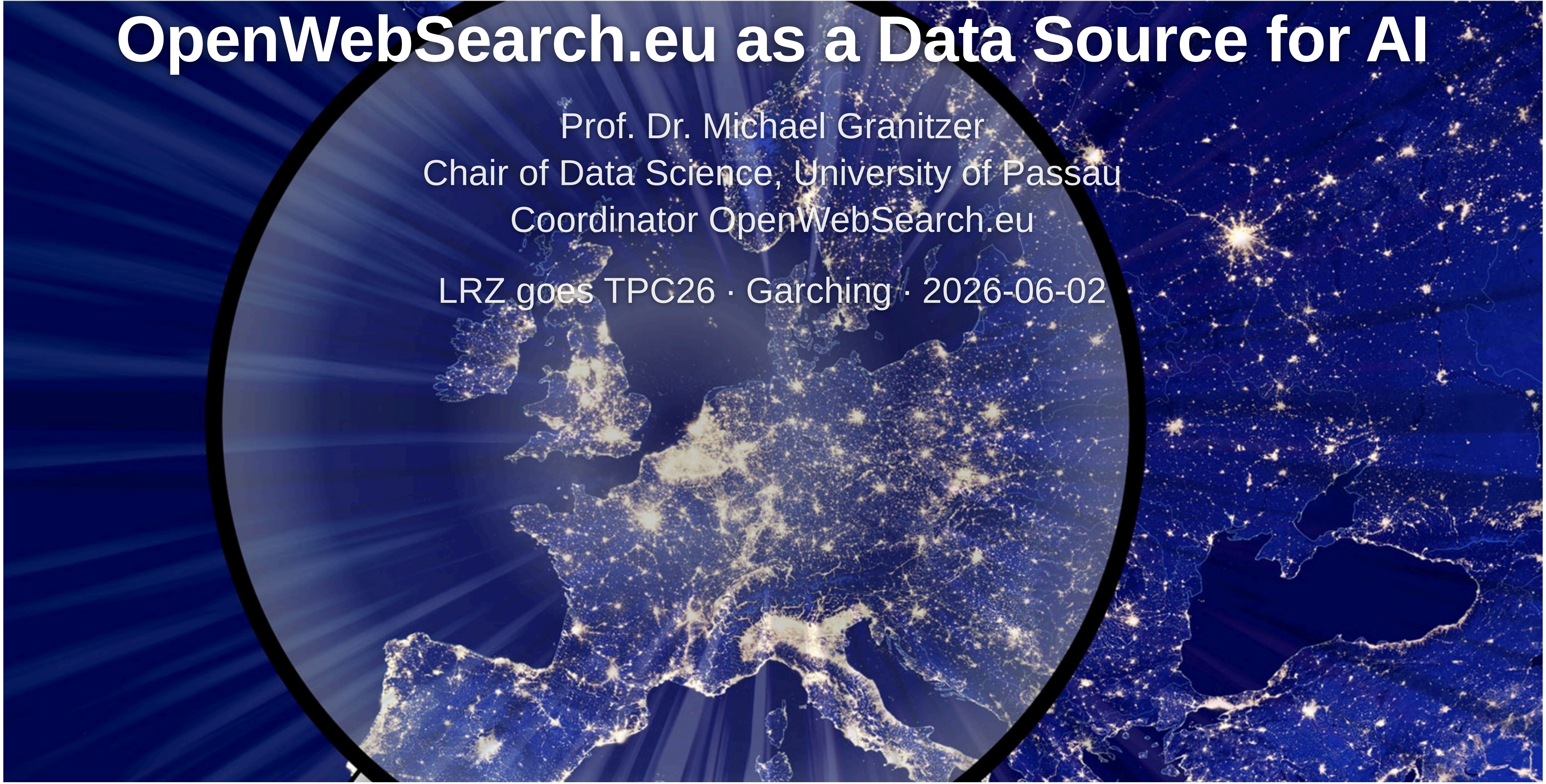
OpenWebSearch.eu as a Data Source for AI



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LRZ goes TPC26 · Garching · 2026-06-02

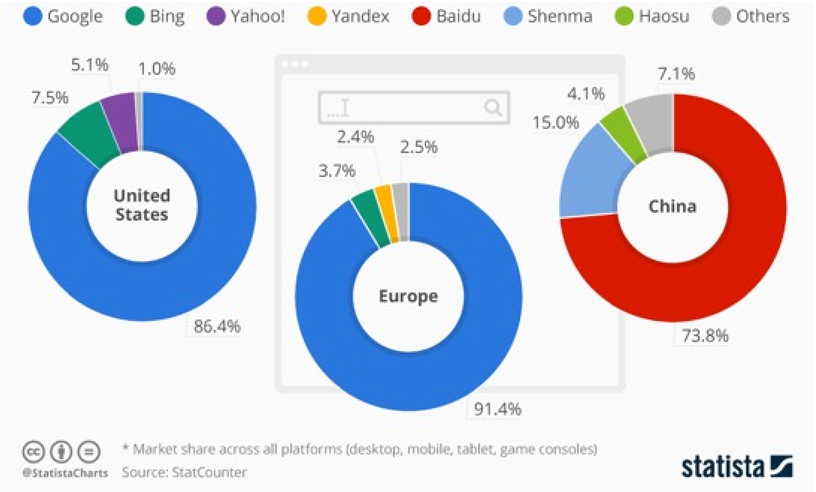
Funded by

Bundesministerium für Forschung, Technologie und Raumfahrt

Bayerisches Staatsministerium für Wirtschaft, Landesentwicklung und Energie

Web Search & Web Data - a Critical Infrastructure

- A **critical infrastructure** for society — and now for AI — comparable to satellite navigation
- A **market oligopoly**: controlled by a handful of gatekeepers (Google, Microsoft, Baidu, …)



Consequences

- **Reduced user choice**
- **Dependence in the information space**
- **Smaller researches and innnovators are locked out**

How to enable researchers, innovators and smaller companies to utilise the web as **data resource** for training and fine-tuning AI systems and to develop **AI-based Search Systems** without massive investments in infrastructure?

OpenWebSearch.eu

> Building an **Open Index of the Web** to lower the cost for researchers, innovators, and companies to **tap the Web as ressource**

Four objectives

1. **Open technology stack** — open pipelines for crawling, preprocessing, indexing
2. **Resource provision** — a network of providers on EuroHPC / AI Factories
3. **Added-value services** — dashboard, tools, data-access APIs
4. **Bootstrapping the ecosystem** — third-party calls, community, use-cases in **search, analytics, and AI**.

14 core partners

Infrastructure

Research

NGOs

Businesses

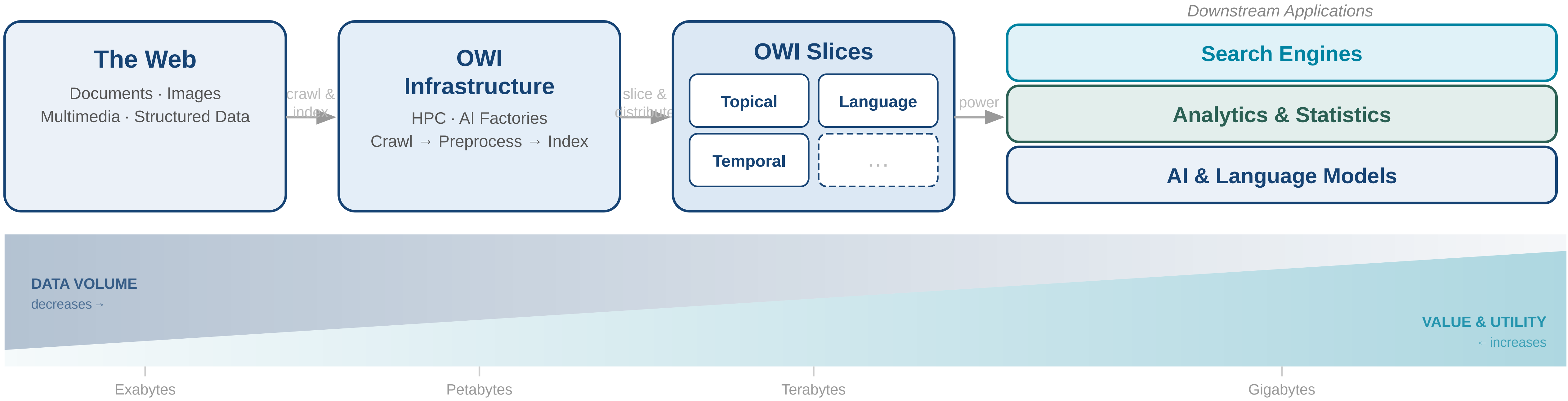
Uni Passau · LRZ · CERN · CSC · IT4I · DLR · TU Graz · Bauhaus-Uni Weimar · Radboud · Uni Leipzig · A1 Telekom · Open Search Foundation · NLnet · SUMA-EV — plus 16 funded third-party projects.

Funded by Horizon Europe (Next Generation Internet programme).

The Goal: An Open Web Index

A **web index** is the data structure behind every search engine — fast, query-based access and ranking over web documents.

> **Our proposition:** a collaboratively created, **open and transparent** web index, running on Europe's HPC centres and AI Factories.



Building the Index

Federated HPC Workflow

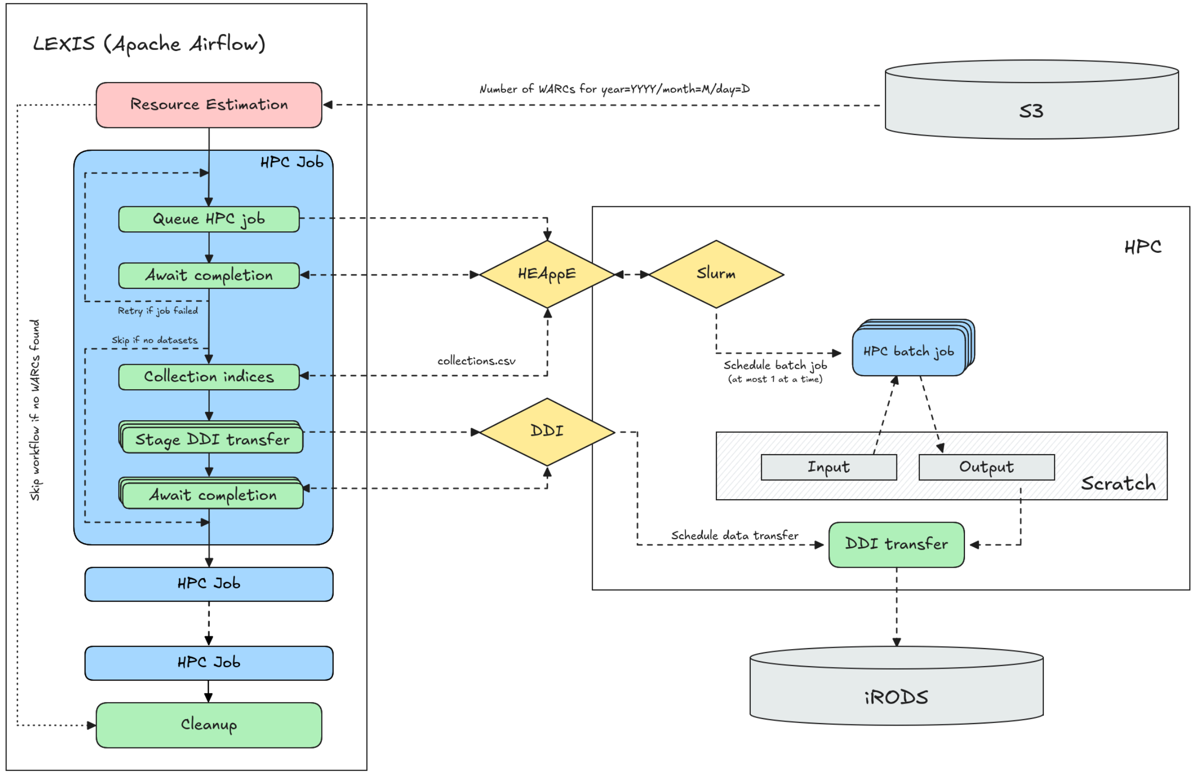
- Continuous Crawling on Cloud Partitions
- Daily Batch HPC Workflows: WARC processing · indexing · semantic enrichment, running
- Runs federated at **LRZ** (Garching) · **IT4I** (Ostrava) · **CSC** (Helsinki)

Federated storage

- Object Storage for Crawl Data
- File Storage for Index Provision
- SQL and slow full-text search for file storage based slice and dice

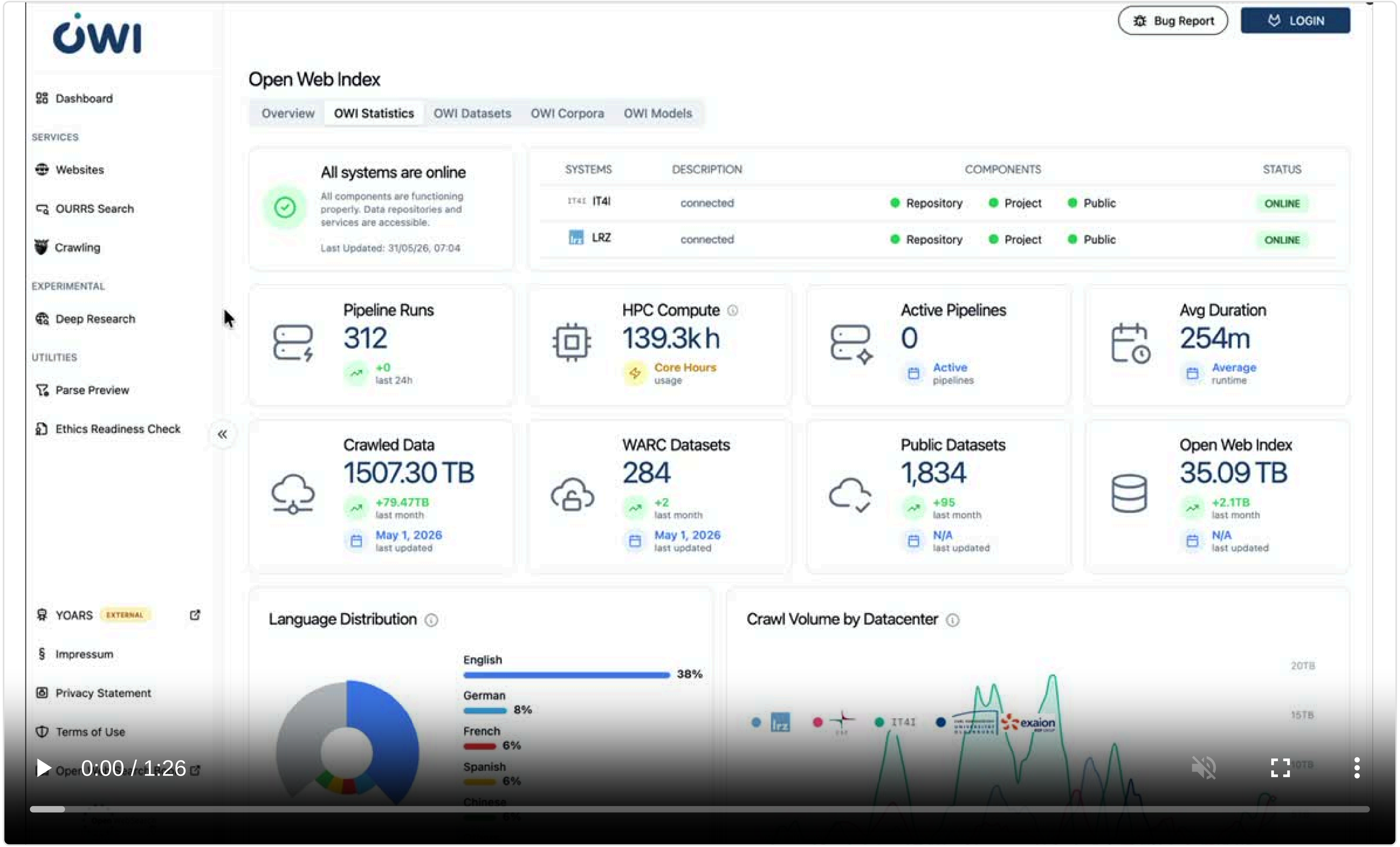
Orchestration

- LEXIS (Apache Airflow) orchestrates jobs across schedulers
- stages data transfers, aligns daily output across all centers



LEXIS / Apache Airflow: federated OWI pipeline orchestration

The Open Web Index Today



1,507 TB
data crawled

35 TB
Open Web Index

1,834
public datasets

284
WARC datasets

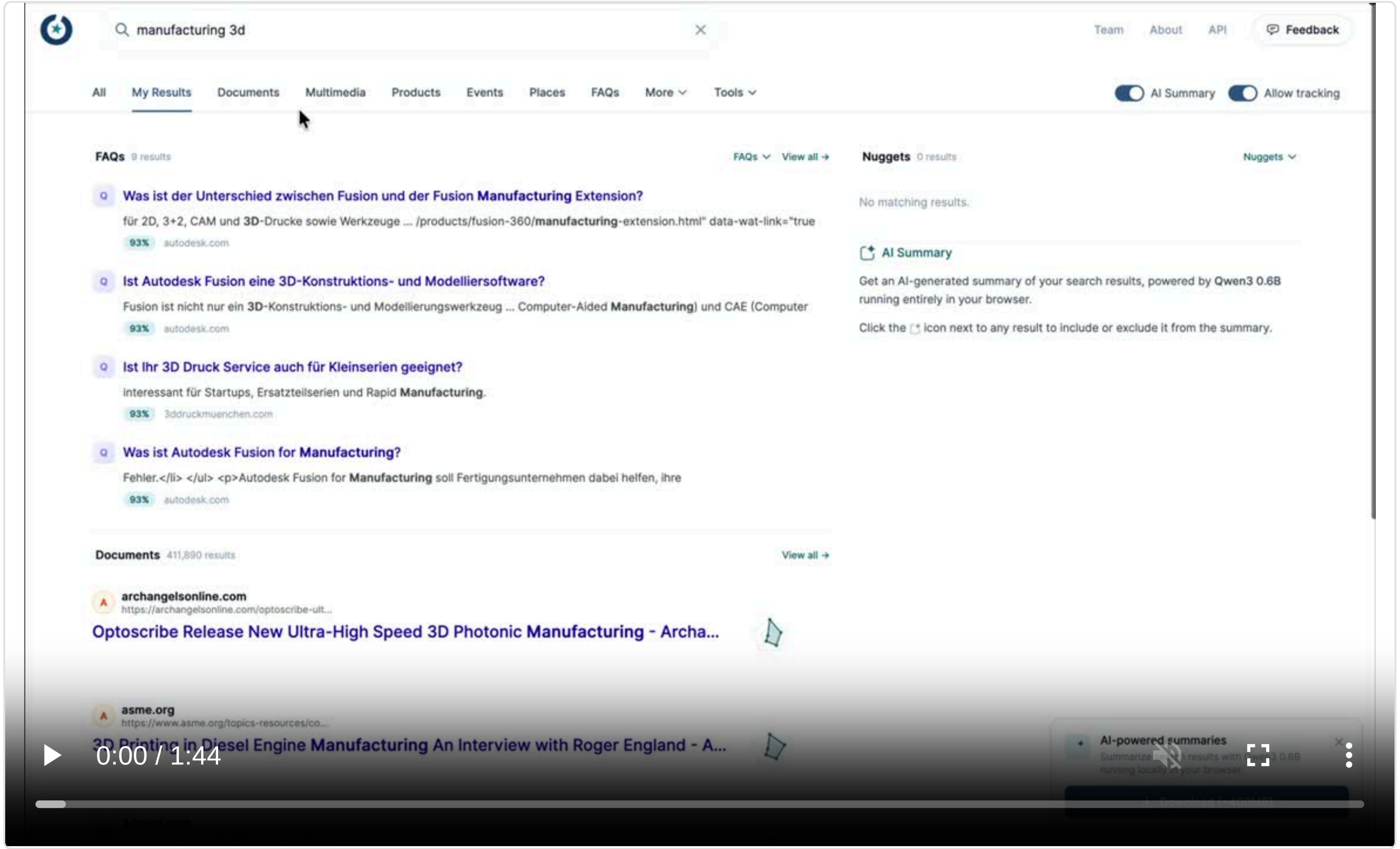
Continuously refreshed — **+79 TB crawled and +95 datasets in the last month** alone.

▶ Statistics 0:10 ▶ Datasets 0:35 ▶ Corpora 0:51Inin ▶ Models 1:14

OWI dashboard — live operational statistics. http://openwebindex.eu

OWI dashboard, figures as of May 2026.

OUR^2S Web Search Engine



2 Months of data

04/25 + 05/25

598.1M Pages

453.4M Entities

14 verticals

documents · media · products · events · places · FAQs · Scholarly · Artefacts · Actions · Pages · Sites · Persons · Conversations · Courses

in-browser AI

private Qwen3 summaries, ~400 MB

OUR²S turns the OWI into a **working search engine** — documents *and* structured entities, summarised privately in the browser.

Web data for training & fine-tuning

Modalities: The Index as a Multimodal Filter

MEDIA ASSETS IN ONE MONTH OF THE OPEN WEB INDEX

600M
web documents

142.1M
media assets

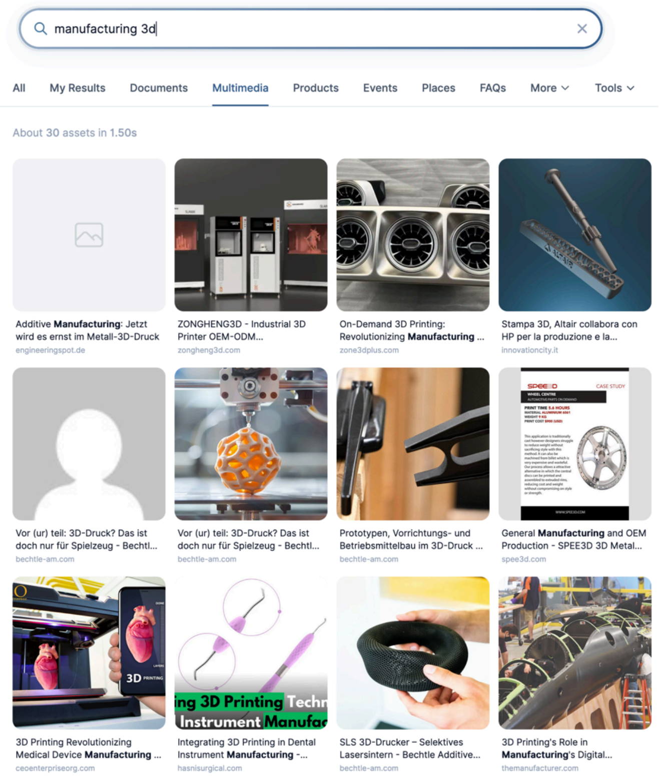
127.8M
images

4.9M
videos

532k
other media

208k
audio

The index lets you **select training data** by modality, language, quality, and host — *before* paying to download and process it. Storage costs drop when re-downloading text & multimedia can be **targeted and selective**.



Example: multimedia results for a manufacturing query — images, 3D models, product shots.

Asset counts by schema.org type from the Open Web Index API (api.ourrs), one-month corpus, queried 2026-05-30.

Domain-Specific Data

"MANUFACTURING" DOCUMENTS IN A SINGLE MONTH OF THE OPEN WEB INDEX

6.3M	2.5M	996k	583k	498k
"manufacturing"	"CAD"	"spare parts"	"3D printing"	"datasheet"

Also present: CNC machining (150k) · injection molding (120k) · industrial automation (562k) · robotics (788k) · PLC (782k) — a domain corpus you can filter out of the OWI.

Keyword matches over one month, English-leaning — a **candidate pool** before curation, dedup, and licensing.

Manufacturing is **just one example** — the OWI holds rich, filterable corpora for *any* domain (law, health, science, finance, engineering, …), **millions of documents per month**.

Document counts from the Open Web Index API (api.ourrs), one-month corpus, queried 2026-05-31.

Structured Web Data

OWI parses **schema.org / Microdata** and classifies pages — including **forums and Q&A** — into **453.4M typed entities**, with three kinds of value:

~83M Deeper knowledge

articles (79.4M) · scholarly (897k) · books (1.7M) · datasets · courses · medical

→ **training & RAG**

~39M Behavioural

blog posts (18.2M) · questions (17.5M) · Q&A pages (717k) · reviews · social posts · forums

→ **instruction & evaluation**

~120M Search & agents

products (91M) · persons (10M) · organisations (6.2M) · events (5.2M) · search actions (3.7M)

→ **agentic search & affordances**

The same crawl yields **knowledge to learn from, behaviour to learn from, and entities to act on**.

Search becomes the memory layer

What lives in the weights, and what stays retrievable?

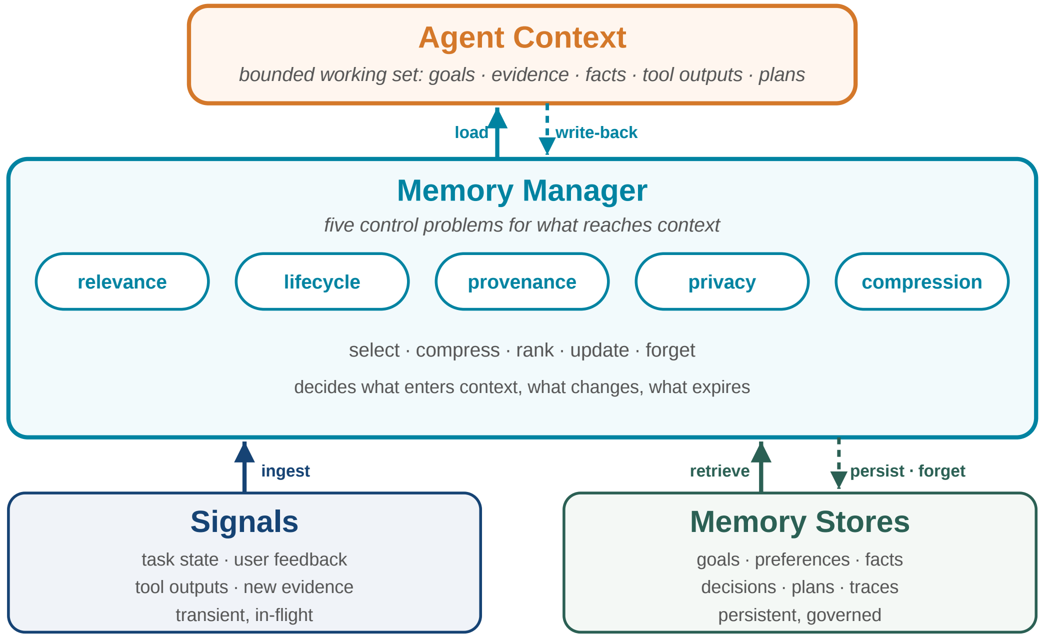
Search as Memory: Parametric vs. Non-Parametric

Retrieval-Augmented Generation (RAG)

Search for context relevant documents.
Usually as Vector search, to cover linguistic and semantic diversity.

RAG introduces a "memory" hierarchy:

- **parametric:** model capabilties & internal knowledge
- **non-parametric:** external knowledge



> RAG is essential for freshness, domain specific adaption and data privacy, but requires scalable and efficient **search infrastructures** in an AI system

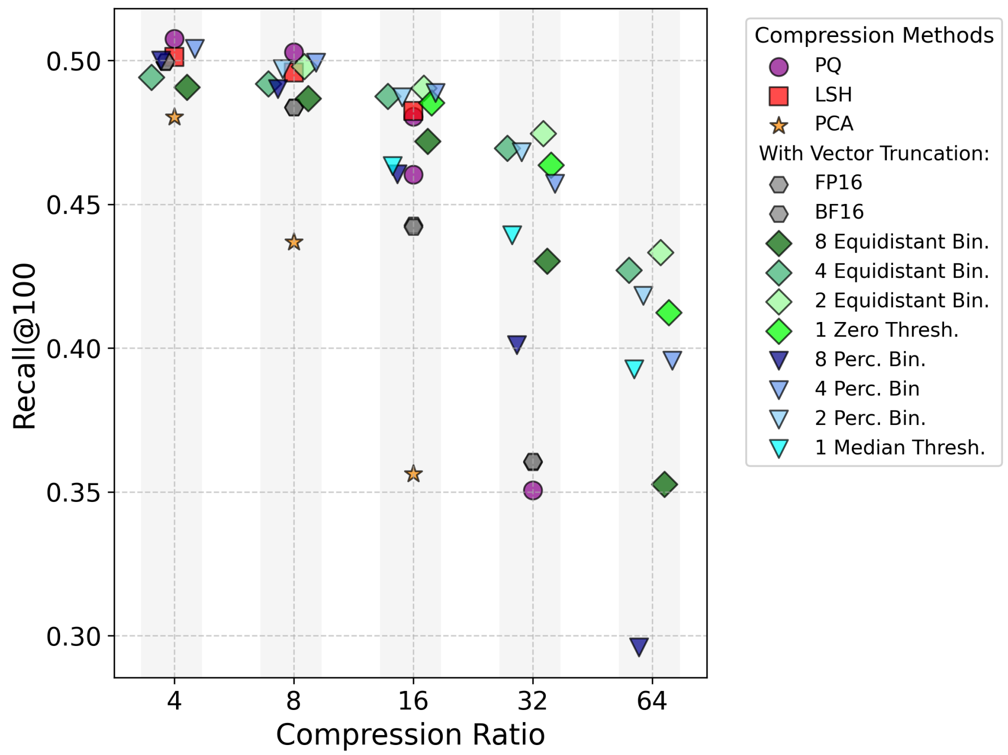
Lewis et al. 2020, Retrieval-Augmented Generation (arXiv:2005.11401); parametric vs. non-parametric memory, Mallen et al. 2023.

Web-Scale Vector Memory Has a Cost

The Open Web Index provides **embeddings** for retrieval on selected web documents, but increases compute and storage significantly.

Why compression matters

- Compute is managable / scalable
- But embeddings are **expensive to store and serve** *($O(n)$ for exact queries)*
- compression trades index size against retrieval quality
- Quantisation wins: results hold recall while compressing **16–32×**



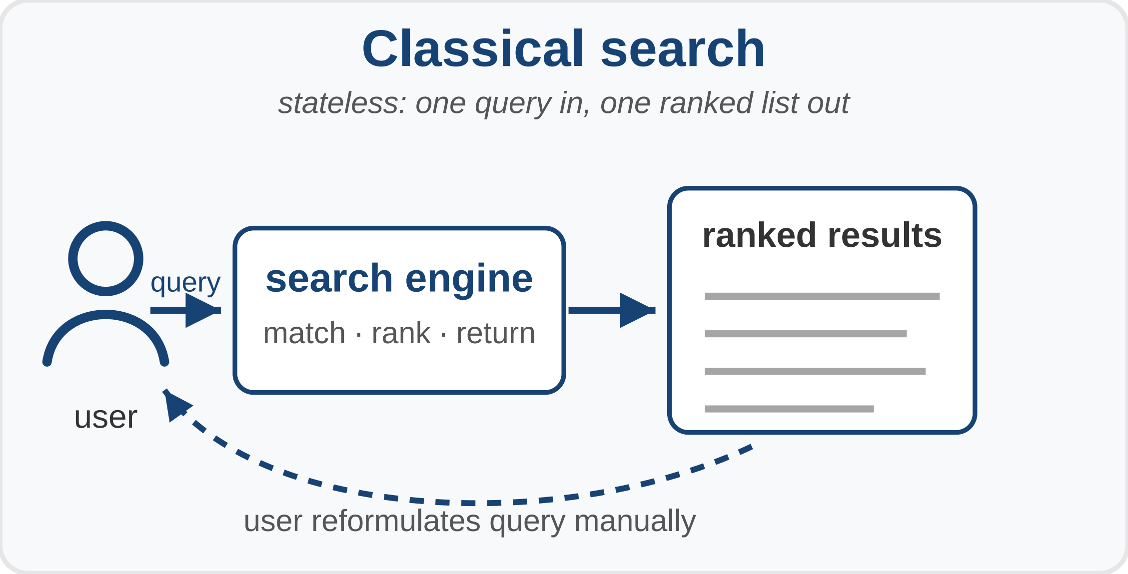
> Compression essential for **scaling up search**.

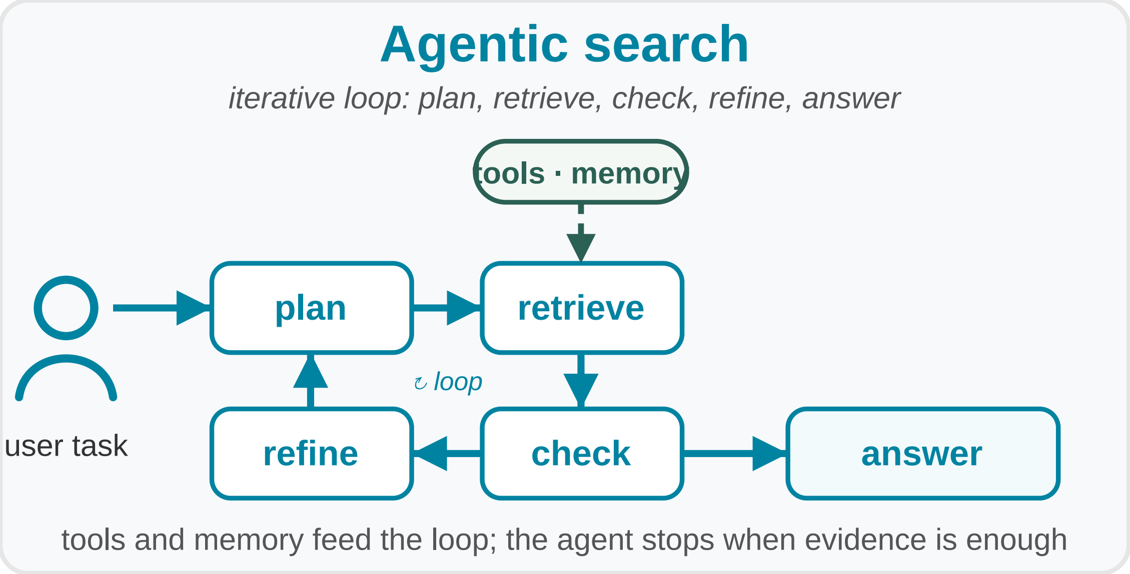
*Own work — **CoRECT**: Controlled Retrieval Evaluation of Compression Techniques (Jina v3, Recall@100 vs. compression ratio). arXiv:2510.19340 · padas-lab-de.github.io/CoRECT*

From retrieval to agents

Retrieval moves into a control loop — and needs actions and traces.

Agentic Search: Retrieval in a Control Loop

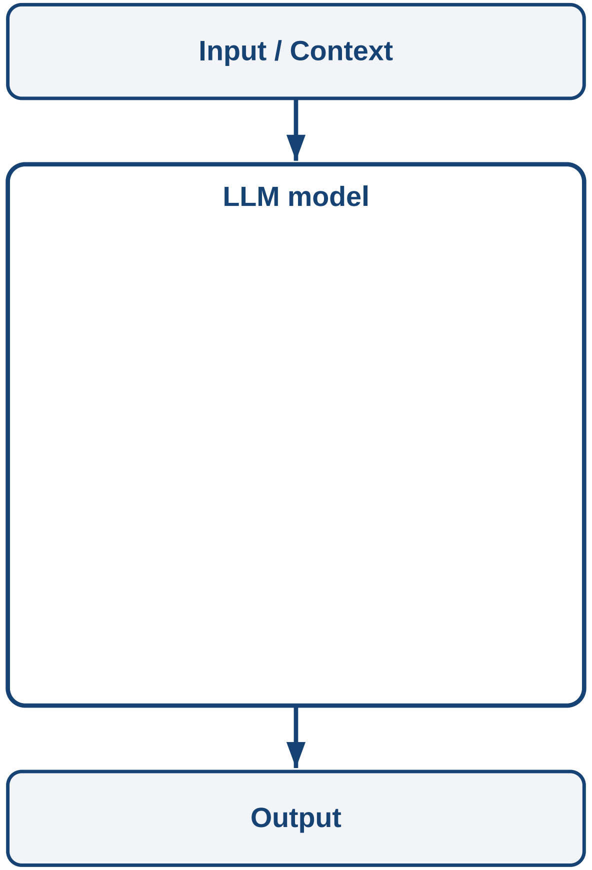




Agents do not just **consume** search results — they **control** retrieval: plan, inspect evidence, call tools, reformulate, synthesize with provenance.

Related paradigms: ReAct (arXiv:2210.03629); Toolformer (arXiv:2302.04761); Self-RAG — adaptive, self-reflective retrieval (arXiv:2310.11511).

A System View of Agentic Search



How much knowledge must sit in **weights** vs. **external memory**? Externalising it → **smaller model + fresher, controllable knowledge** — trading latency & more loops for

flexibility, control. *Where to invest resources?*

Memory roles after CoALA: working · episodic · semantic · procedural (Sumers et al. 2023, arXiv:2309.02427).

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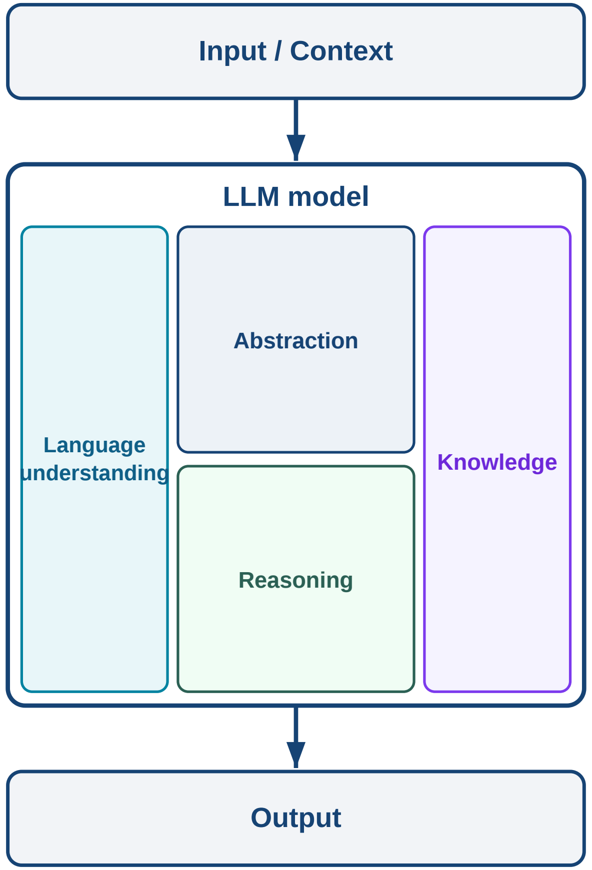

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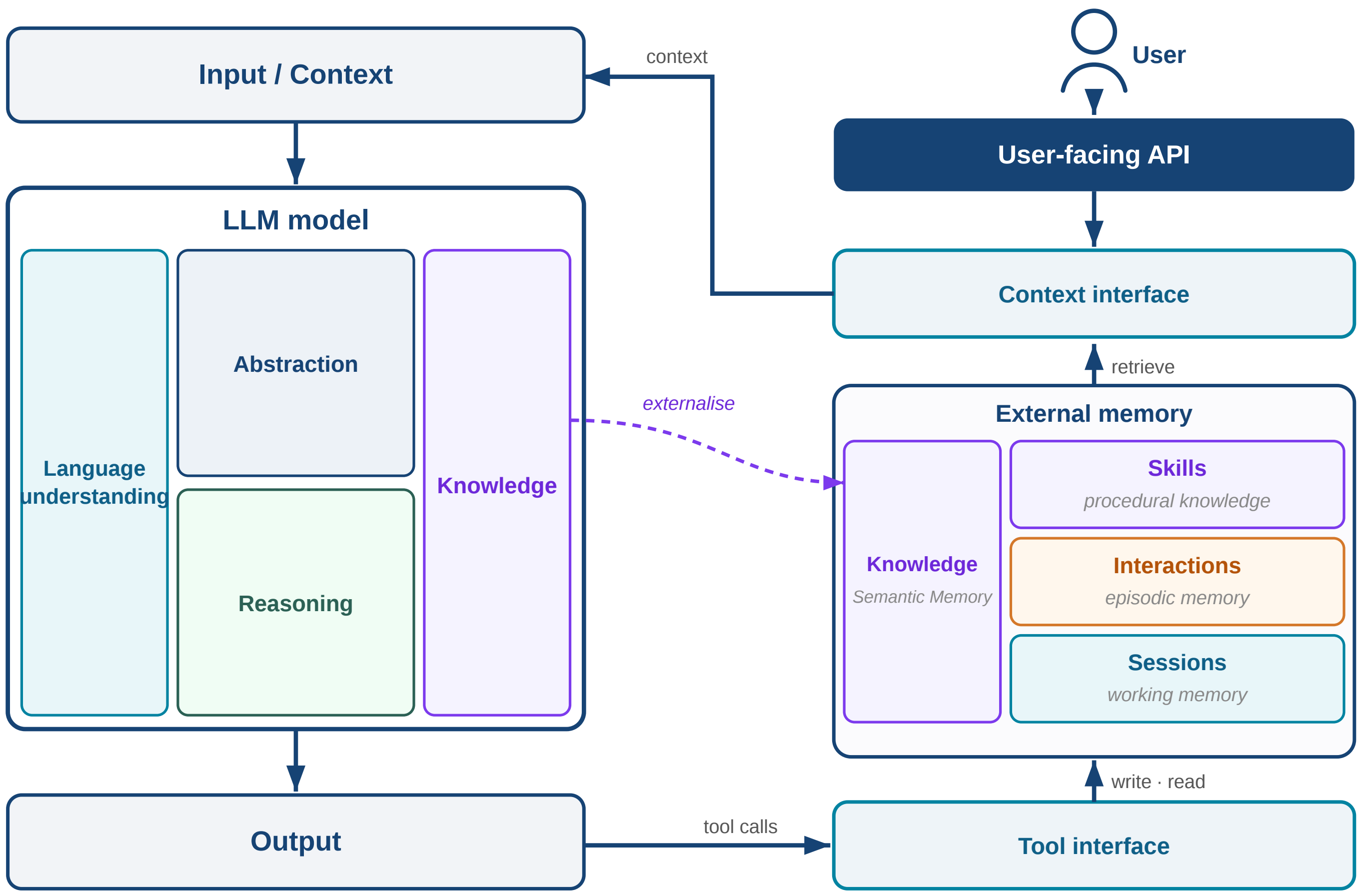

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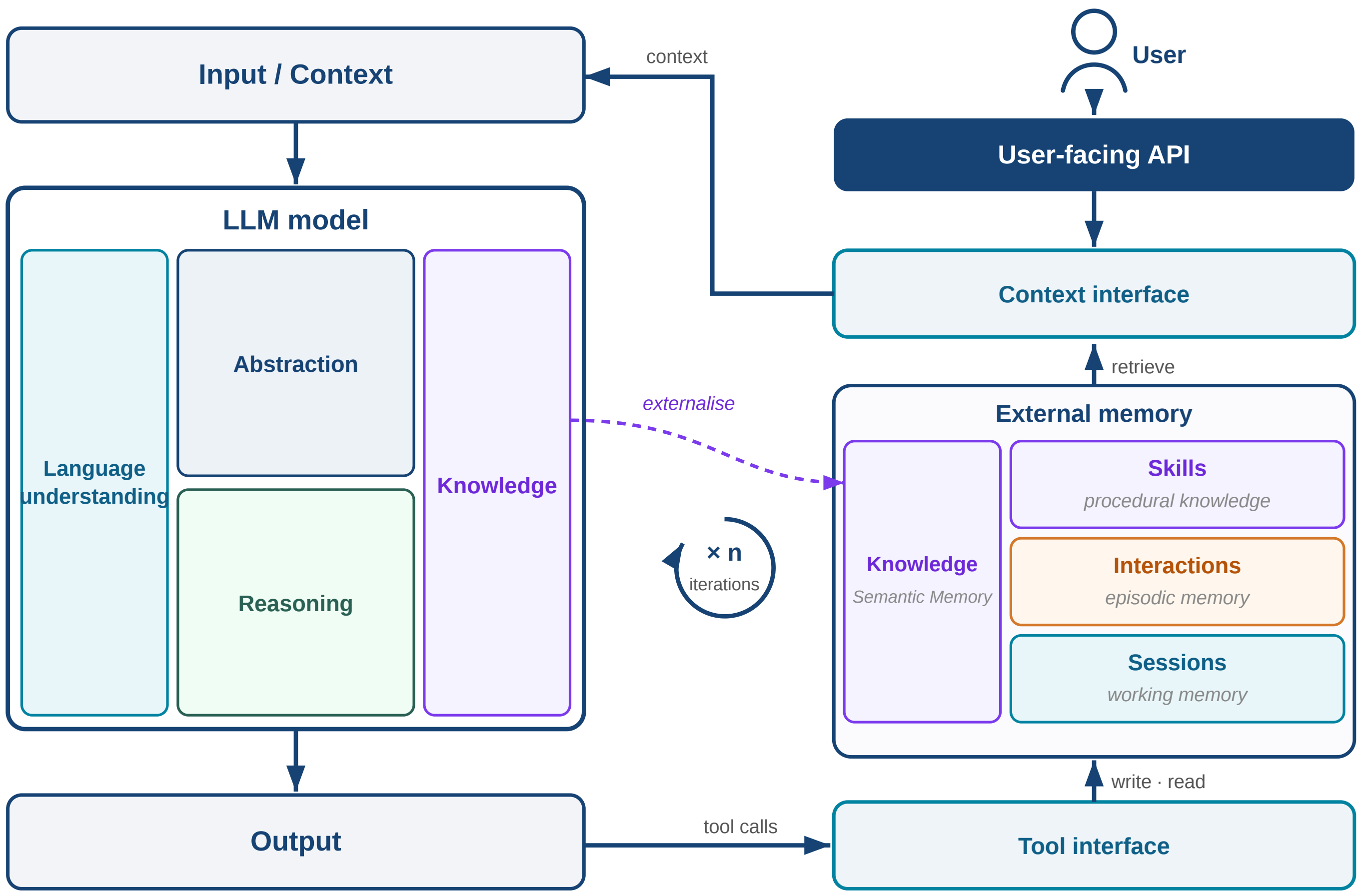

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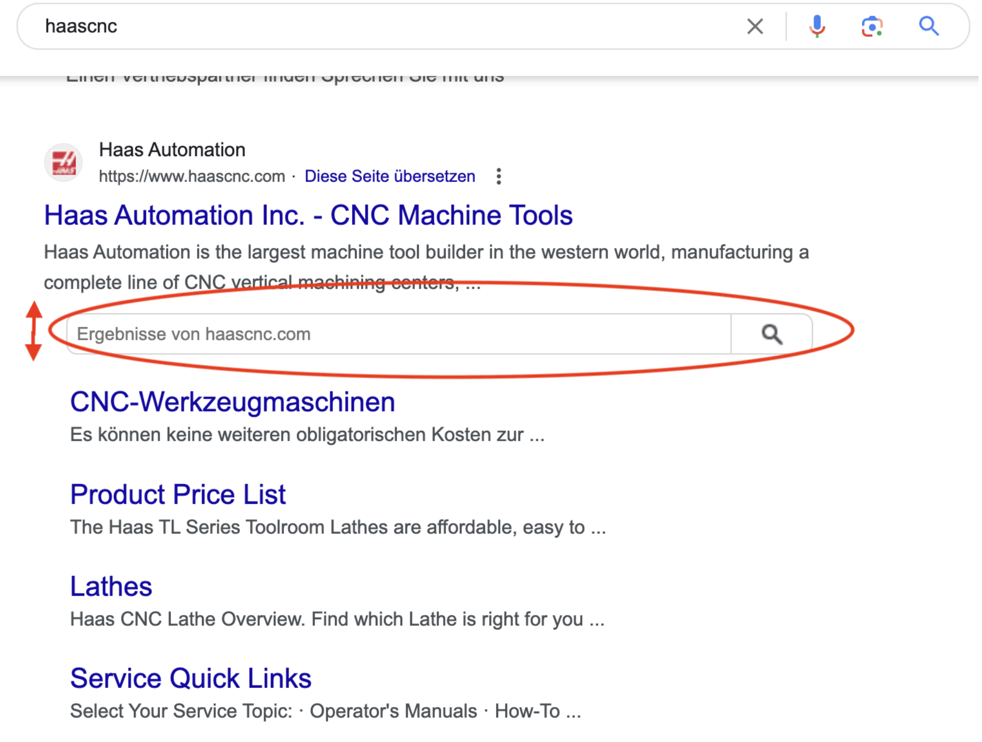
Structured Web Data Becomes Agent Affordance

ACTION & APPLICATION ENTITIES IN THE OPEN WEB INDEX

3.65M SearchAction	**909k** SoftwareApplication
114k WebApplication	**94k** MobileApplication

> Agents need **discoverable actions** — what can be searched, called, installed, or configured. The index is a **candidate generator**, not blind trust.



*Sitelinks searchbox = schema.org **SearchAction**: agents can call a site's own search. Source: metayer.de*

Entity counts by schema.org primary_type, Open Web Index API (api.ourrs), corpus 2026-05.

Beyond Web Data: Traces & Evaluation

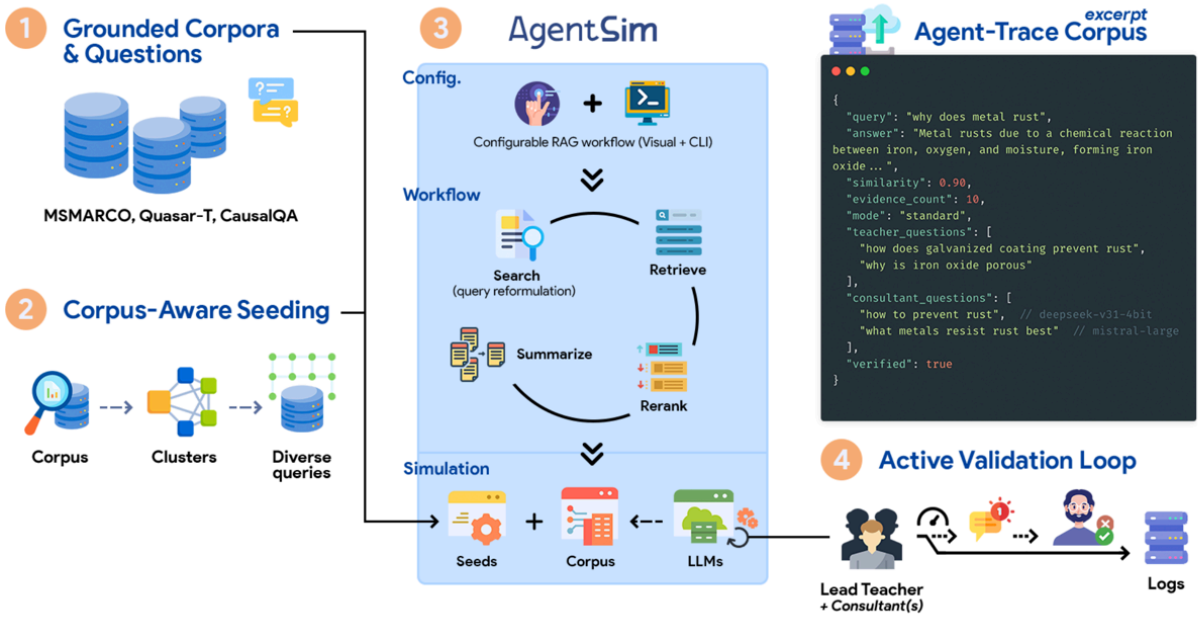
OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

Synthetically creating agent traces

AgentSim — simulates RAG agents over document collections to generate synthetic, **document-grounded** reasoning traces

- **103k+** verifiable steps (Agent-Trace Corpus)
- 100% grounding rate on substantive answers.
- Agent-trace synthetic data generating is both **GPU compute heavy** and requires **search systems**.

AgentSim, Agent-Trace Corpus · arXiv:2604.26653



Comparing to simulated human users

UXSim — grounds an adaptive LLM agent with classical simulators → accurate, **explainable** user-search simulation without costly user studies.

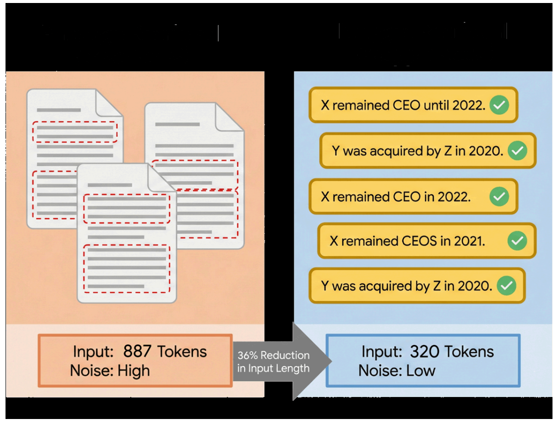
Beyond Web Data: Knowledge & Interaction

OPEN CHALLENGES IN AGENTIC SEARCH — OUR WORK

How to structure knowledge so agents find it?

NuggetIndex — distil documents into atomic **"nuggets"**: 887 → 320 tokens, far less noise, facts kept for grounding.

Own work — nugget-level indexing for agent knowledge



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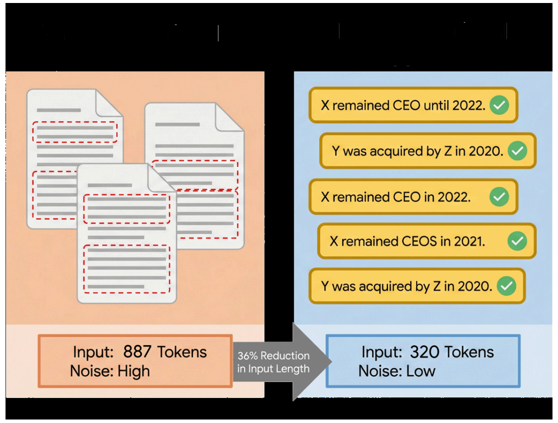
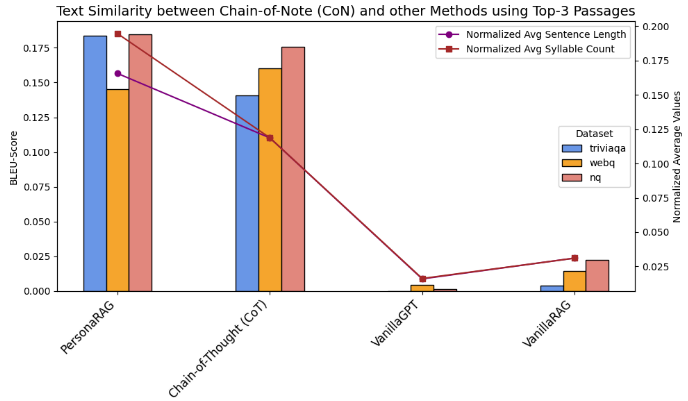


How should retrieval adapt to the user?

PersonaRAG — user-centric agents personalise retrieval → higher answer quality (BLEU) than CoT and vanilla RAG across TriviaQA, WebQ, NQ.

Zerhoudi & Granitzer 2024 · arXiv:2407.09394



Conclusion

Tokens
pretraining
→
Examples
fine-tuning
→
Passages
RAG memory
→
Actions
tools & APIs
→
Traces
behaviour & feedback

What we covered

- The web is an open, reusable **data layer** — OpenWebSearch delivers **data + an index**
- **Filter** it for training & fine-tuning; **serve** it as RAG memory & agentic search
- Agentic search needs more than data: **interaction, knowledge structure, behaviour, evaluation**
- Developing Agentic Search is not only GPU-intensive, it requires complex systems.

Let's cooperate

We are **looking forward to collaborations** on open web data, retrieval memory, and agentic search — research projects, datasets, and use-cases all welcome.
→ openwebsearch.eu · Chair of Data Science, University of Passau

References

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